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From Classroom to Enterprise: Architectural Patterns for LLM Integration

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Two Organizations, Similar Problem

The Specification Gap

Academic

- AI discouraged
- Behavioral restriction
- Weak workflow specification

Enterprise

- AI encouraged or mandated
- Behavioral expectation
- Weak workflow specification

Similar structural drift

Weak architecture emerges where policy is unworkable.

LLMs as Active Components

Passive Tool vs. Active Component

Passive Tool

- Fixed function
- Deterministic behavior
- No adaptive logic generation
- Bounded state effects

Active Component

- Generates logic
- Shapes decisions
- Materially affects downstream system state
- Sits inside the control loop

The issue is not whether the model is intelligent. The issue is whether it changes system behavior.

Two Architectures

Control Boundary Location

Centaur

- Gated validation control
- Human retains authority at explicit checkpoints
- Model generates proposals
- Human validates
- Discrete decision boundaries effects

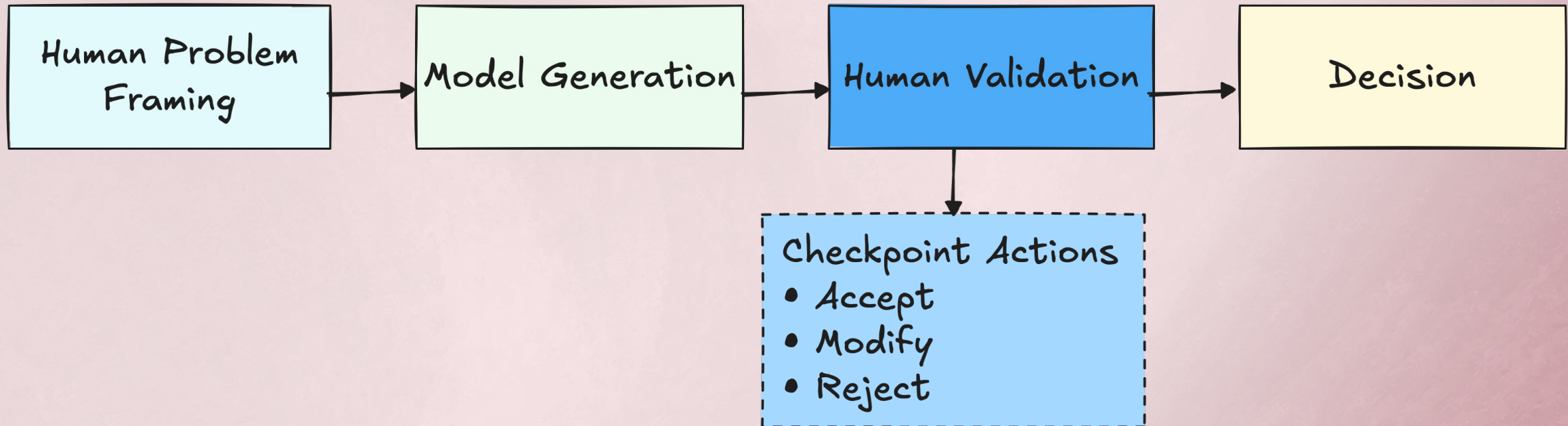
Cyborg

- Supervisory control
- Human monitors and intervenes by exception
- Model operates continuously across the workflow
- Dynamic authority sharing

These are structurally different control architectures. Deployment should follow purpose, risk, and reversibility.

Centaur Architecture

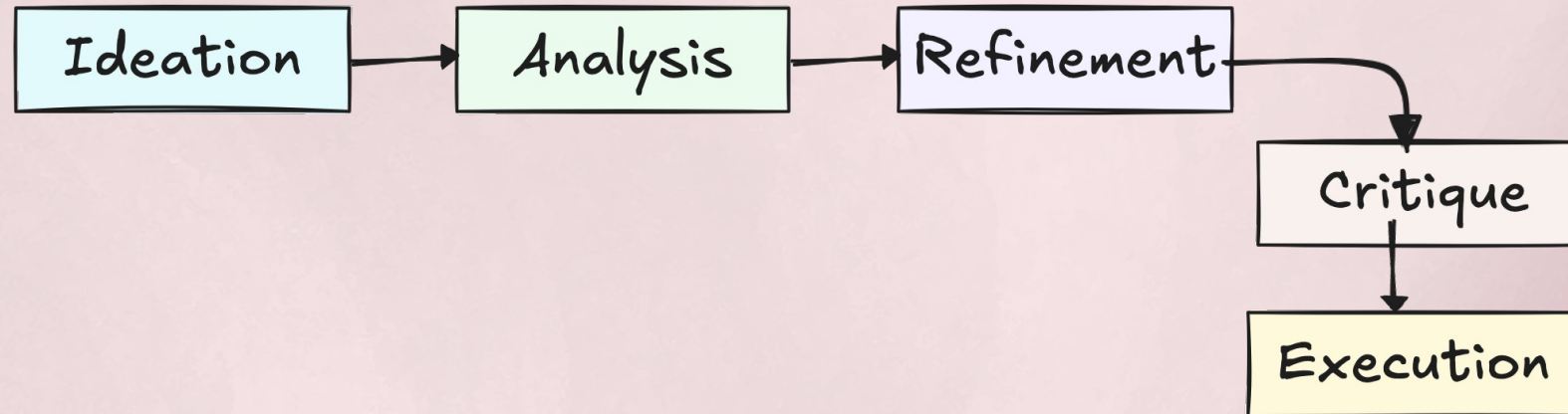
Centaur Architecture -- Explicit Validation Gate



Critical Failure Mode: Degraded oversight
Review remains in form, weakens in substance.

Cyborg Architecture

Cyborg Architecture -- Shared Process



Human Role

- Monitors
- Guides
- Intervenes by Exception

Failure Modes

- Observability Gaps
- Error Propagation
- Responsibility Diffusion

Cyborg workflows can be powerful,
but they require mature control environments.

What I Observed

Opposite Governance, Similar Drift

Academic

- 7 semesters
- ~300 graduate students
- AI use discouraged or constrained

Enterprise

- Mid-size technical organization
- ~40 engineers and operations staff
- AI encouraged and later mandated

Explicit validation became ceremonial.
Similar degradation across opposite policy regimes.

Three Phase Evolution

Three Phase Evolution -- Adoption Progression

Phase 1: Atomic Use

- Bounded Tasks
- High Scrutiny
- Low Reliance

Phase 2: Delegation

- Expanding Use
- Nominal Validation
- Danger Zone

Phase 3: Differentiation

- Some remain atomic
- Some stabilize into stronger centaur patterns
- Some drift into degraded review
- Small minority move into cyborg workflows

Risk concentrates in the middle, where oversight still appears present but is already weakened.

FMEA: Risk Assessment

FMEA -- Risk Assessment
Detectability degrades across phases

Assessment Lens

- Severity
- Occurrence
- Detectability

Key Pattern

- Phase 1 --> Failure Easier to Detect
- Phase 2 --> Detectability Weakens
- Phase 3 --> Detectability very low

Core Risk

- Authority drift
- Validation erosion
- Polished artifacts masking degraded control

The most dangerous failures are often the hardest to see while they are forming.

Architecture Selection

Architecture Selection -- Consequence Driven

Wrong Question:

What can the model do?

Better Question:

What happens when the workflow is wrong?

Selection Logic

High-impact / irreversible / compliance-sensitive --> explicit validation gates, strong traceability

Lower-impact / reversible --> looser structures possible

Architecture should follow consequence. Do not skip stages.

What This Means

What This Means Practical Implications

What Fails

- Mandated usage without workflow design
- Blanket prohibition without alternatives
- Premature Cyborg deployment

What Is Needed

- Verification capacity
- Enforced control points
- Traceability embedded in workflow

Governance Tools

- Potentially useful
- Insufficient if overlays

The deeper issue is architectural, not merely behavioral.

What This Paper Contributes

What This Paper Contributes Three Contributions

1. Architectural Taxonomy

- Collaboration patterns as systems engineering framework

2. Convergence Evidence

- Similar degradation patterns across opposing governance contexts

3. Exploratory FMEA

- Architectural choices linked to distinct failure modes
- Basis for continued research

Treat Human-LLM collaboration as an architectural design problem.

Open Questions

Open Questions -- Research Agenda

- What predicts architectural degradation?
- What preserves real control?
- Which governance mechanisms hold under operational pressure?
- How do we lower the stakes when supervision weakens?
- How should governance evolve when governance itself requires increased automation?

The durable solution is not perfect users. It is more resilient systems.